BUS-F420: Equity and Fixed Income Investments

Lecture 15: Interest Rate Risk + Intro to the Yield Curve

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March 23, 2026

Recap from last lecture

Goal: Understand how interest rates affect bond prices

- Bond prices respond to changes in market yields $\rightarrow$ **interest rate risk**
- We need tools to measure this sensitivity $\rightarrow$ **Duration!**
- **Today:** Review basics of duration, spend more time on caveats

6 rules for interest rate sensitivity:

General Rules:

1. Bond prices and yields are inversely related ($y \downarrow \Rightarrow P \uparrow$)
2. Bond prices change less when yields rise as oppose to fall ("convexity")

Maturity [A $\rightarrow$ B]:

3. Long maturity bonds are more sensitive to yields than short maturity bonds
4. Price sensitivity is increasing but at a decreasing rate as maturity increases

Coupons [B $\rightarrow$ C] & Yields [C $\rightarrow$ D]:

5. Low coupon bonds are more sensitive to interest rates than high coupon bonds
6. Price sensitivity is smaller at high yields

Duration for a General Option-Free Bond

- For an option-free bond with cash flows CF_t, **(modified) duration** is defined as

$$\boldsymbol{D} = \frac{1}{1+y}\sum_{t=1}^{T} \boldsymbol{w_t t}, \qquad \boldsymbol{w_t} = \frac{\boldsymbol{CF_t}}{(1+y)^t P}$$

- Duration takes each cash flow's maturity into account, which implies (all else equal)
  - Longer maturity bonds have *higher* duration
  - Higher coupon bonds have *lower* duration

How is Duration Related to Interest Rate Risk?

- For a general bond with generic cash flows, the price effect of a **small** change in yield is

$$\frac{\Delta P}{P} = -\boldsymbol{D} \times \Delta y$$

- **Duration** allows us to <u>combine</u> all aspects of a bond into a single measure

Implications

- Two bonds with identical time to maturity can have different duration
- Two bonds with identical coupons can have different duration
- Two bonds with identical time to maturity *and* coupons can have different duration

Three Caveats with Duration

1. Only holds exactly for *small* changes in rates
   - **Convexity**

2. Only holds exactly for *option-free* bonds
   - **Negative convexity, effective duration**

3. Only holds exactly for *constant yields*
   - **Yield curve $\rightarrow$ today + Wednesday**

Convexity

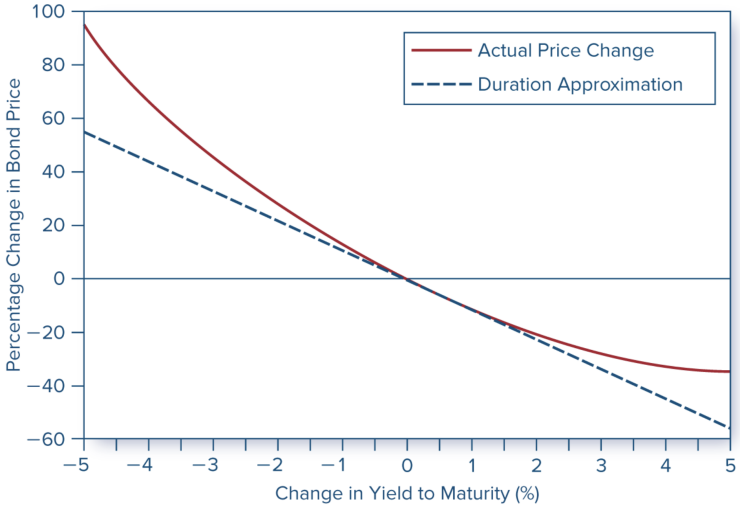
Caveat 1: What if Interest Rate Changes Are Large?

- There's a reason I've emphasized **small** changes in interest rates: it's only an approximation

- With long maturities, bond prices are highly **non-linear** in yields
  - Reduces the accuracy of our duration formula
  - (Think about calculus: a derivative is a *local* concept)

- **Previous exercise** highlights the importance of accounting for this non-linearity
  - Call this **convexity**

Visualizing Convexity and Duration Approximation Error



A Revised (But More Complicated) Interest Rate Risk Equation

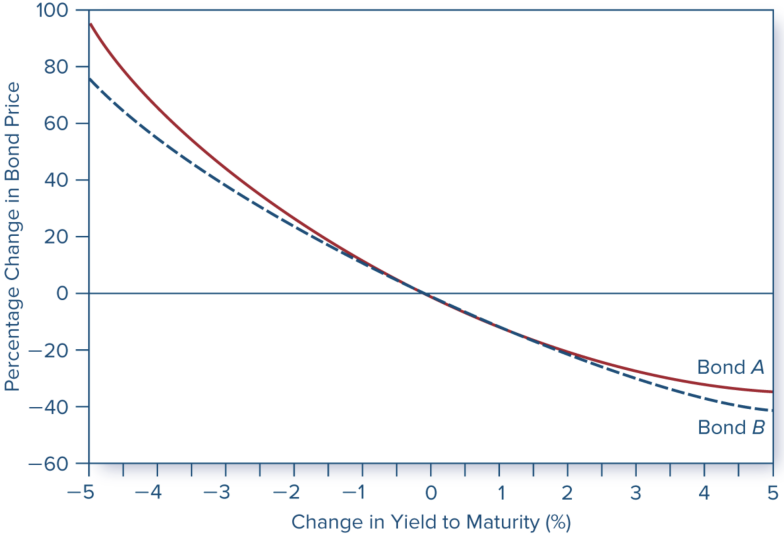
- We can adapt our previous equation to account for convexity:

$$\frac{\Delta P}{P} = -\boldsymbol{D} \times \frac{\boldsymbol{\Delta y}}{\boldsymbol{1+y}} + \frac{\boldsymbol{1}}{\boldsymbol{2}} \times \textbf{Convexity} \times (\boldsymbol{\Delta y})^2$$

- **Convexity** has a very complicated formula that I won't show here

- Why do we add in the term for convexity this way?
  - **Answer:** $\sim$ 2nd order Taylor approximation
  - When Δy is very small, the term basically goes away

Visualizing Convexity



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  1. *Increases* in yields lead to *smaller declines* in bond prices
  2. *Decreases* in yields lead to *higher gains* in bond prices

- Bonds with higher convexity therefore have **higher expected returns**

- **What do you think this implies for bond prices and yields?**

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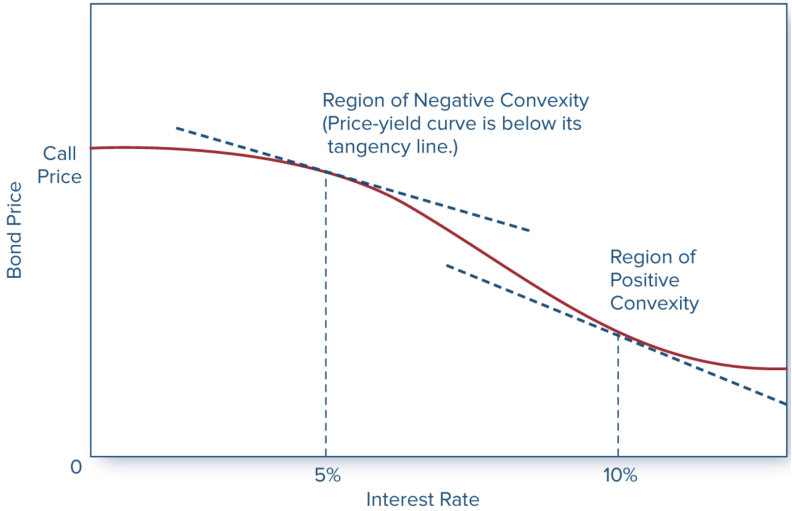
- **What do you think this implies for bond prices and yields?**
  - Answer: convex bonds tend to be priced higher with lower yields

Callable Bonds

What if we don't have an option-free bond?



How Do We Calculate Duration for Callable Bonds?

- We can't apply our formula from before due to the **embedded call option**
  - Cash flows are more complicated!

- **In practice:** directly calculate <u>effective duration</u> using prices and yields:

$$\textbf{Effective Duration} = -\frac{\Delta P/P}{\Delta r}$$

- **Note:** we can't use changes in yields as before! (Yield to maturity unknown)
  - Use *level* shifts in the **term structure** to approximate change in market rate

Model and Duration Quality

- Even if we use effective duration, what assumptions are we making that might be misleading?

Model and Duration Quality

- Even if we use effective duration, what assumptions are we making that might be misleading?
- **A: We assume we know how to determine the price!**
- **Reality:** Callable bond prices often make modeling assumptions, so:
  - If model is bad, duration measure is bad
  - Can have large consequences!

The Yield Curve

Are yields constant over time?

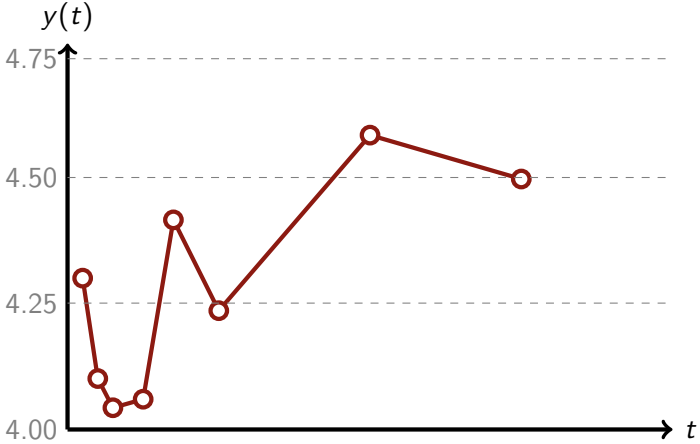
- **Key insight:** longer maturity bonds provide higher yields...
  - ...*conditional on a given bond price!*

- **Reality:** Bond prices vary across maturities! Yields are then inferred from prices.
  - Yields (y_t) across maturities referred to as the **yield curve**

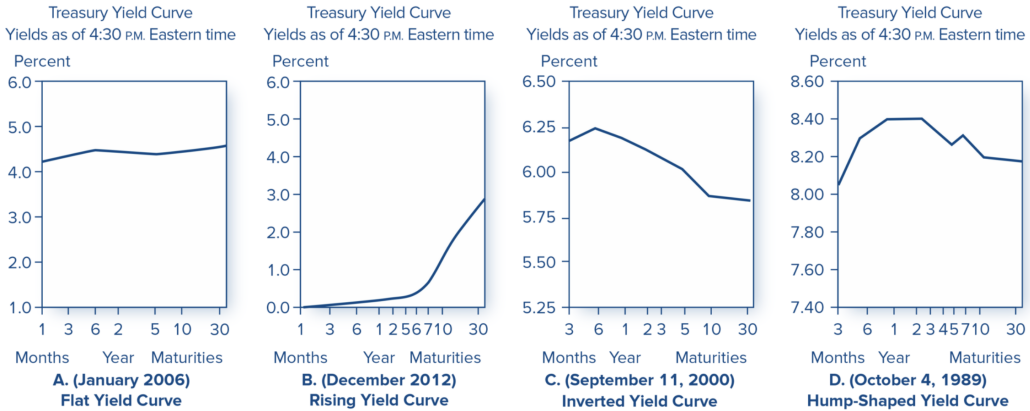
Constructing the Yield Curve For Current Rates

Oct. 28, 2024

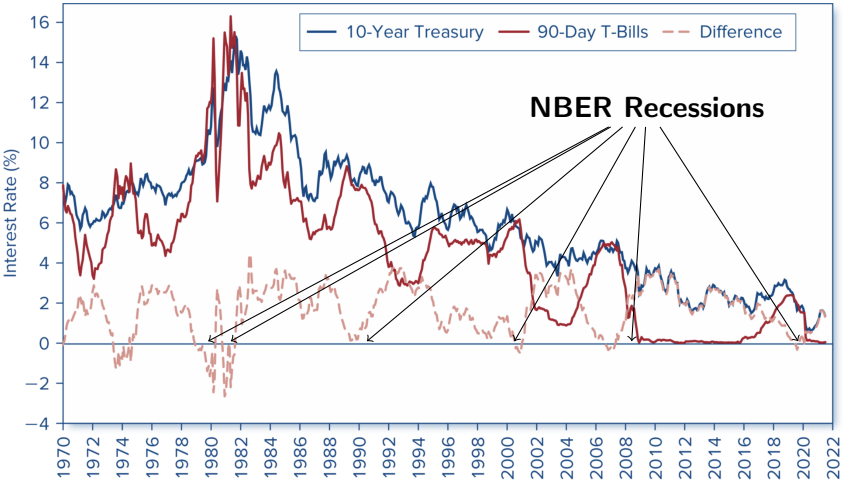
Maturity	Yield
1	4.303
2	4.101
3	4.043
5	4.06
7	4.415
10	4.236
20	4.583
30	4.497



Yield Curves Need Not Be Upward Sloping



The Slope of the Term Structure as an Economic Indicator



Using the Proper Discount Rate for Coupons

- Let's revisit our original valuation equation for a bond:

$$\textbf{Bond Price} = \sum_{t=1}^{T} \frac{\textbf{Coupon}}{(1+r)^{t}} + \frac{\textbf{Par Value}}{(1+r)^{T}}$$

- Remember what discount rates are for: valuing a cash flow *relative to a comparable market return*
  - Different maturities = different market return!
  - Constant discount rate technically incorrect

- **Instead:** Use market rates of each *individual* maturity → "stripping"

Treasury Stripping: How Does it Work?

- **Basic Example:** 2-year treasury paying a \$50 annual coupon and \$1000 par value
  - Strip 1: 1-year zero with face value \$50
  - Strip 2: 2-year zero with face value \$1,050

- Prices should then be equal:

$$\underbrace{\sum_{t=1}^{2} \frac{\$50}{(1+r)^t} + \frac{\$1000}{(1+r)^2}}_{\text{price of coupon bond}} = \underbrace{\frac{\$50}{(1+r)}}_{\text{price of 1-year zero}} + \underbrace{\frac{\$1050}{(1+r)^2}}_{\text{price of 2-year zero}}$$

Using Stripped Treasuries to Value Bonds: Example

- Let's price a bond with the following characteristics:
  - Par Value: \$1000
  - Annual coupon: \$80
  - Maturity: 3 years
  - **Yield Curve:** $y_1 = 5\%$, $y_2 = 6\%$, $y_3 = 7\%$

- **What is the price of the bond? What if we used a constant rate of 7%?**

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$$\text{Price:} = \frac{80}{1.05} + \frac{80}{(1.06)^2} + \frac{80 + 1000}{(1.07)^3} \approx \$1029$$

$$\text{3-Year Price:} = \frac{80}{1.07} + \frac{80}{(1.07)^2} + \frac{80 + 1000}{(1.07)^3} = \$1026.24$$

How does the yield curve affect duration?

- Suppose that the yield curve **steepens** — y_3 now 8%

- **Suppose you calculate prices using the full yield curve. What's the change in price from last slide?**

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Next Time:
Where does the yield curve come from?